

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	1 October 2026
Team ID	6ab7f31f8c7c66e70672939b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Concept Explanation	Student enters a difficult academic concept or topic System processes the requested concept System generates a simplified explanation in clear, student-friendly language System may provide step-by-step clarification where appropriate Model: LaMini-Flan-T5-783M Endpoint: /explain
FR-2	Academic Question & Doubt Clarification	Student enters an academic question System receives and processes the question System generates an appropriate answer System displays the answer clearly to support doubt clarification Model: Google Gemini 3.8 Flash Endpoint: /qa
FR-3	Learning Recommendations	Student provides a topic or learning context System processes the provided topic / context and identifies relevant learning topics System recommends appropriate next topics Recommendations support a logical, structured learning progression Model: Google Gemini 3.8 Flash Endpoint: /learn/recommendations
FR-4	Content Summarization	Student provides academic or learning content System processes the provided content System generates a concise summary that retains the important information Summary is presented in an easy-to-review form Model: Google Gemini 3.8 Flash Endpoint: /summarize
FR-5	Quiz & Knowledge Assessment	Student provides a topic or learning area System processes the topic System generates quiz questions for practice and knowledge assessment Generated questions are presented clearly to the student Model: Google Gemini 3.8 Flash Endpoint: /quiz

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should be simple, clear and student-friendly, with straightforward input and output. Navigation between the learning functions (explanation, Q&A, quiz, summary, recommendations) should be easy, without unnecessary complexity.
NFR-2	Security	Current prototype: API credentials (Gemini API key) are not exposed in frontend code and are kept in environment variables or equivalent configuration, not hard-coded in source files; requests and inputs are validated at the backend. Recommended for production: HTTPS should be used when deployed. No advanced enterprise security is claimed.
NFR-3	Reliability	Valid requests should be processed consistently. Invalid input should be handled appropriately, and failures from the external AI service should be handled gracefully with a clear message, avoiding unnecessary crashes during normal use.
NFR-4	Performance	Performance depends on the lightweight HTML/CSS interface, FastAPI request handling, server-side processing, local LaMini-Flan-T5-783M inference for explanations, Google Gemini API response latency for cloud-based functions, request/response size and concurrent request handling. No performance figures have been measured.
NFR-5	Availability	Current prototype: availability depends on the FastAPI / Uvicorn application running; Gemini-based functions also require the Google Gemini API to be reachable; the local explanation function depends on the local application environment. Possible production improvements (not implemented): multiple application instances, health checks, load balancing.
NFR-6	Scalability	Current prototype: frontend, backend logic and AI services are kept separate, and AI functions are accessed through backend API endpoints. Future growth (not implemented): application instances could be scaled in a production deployment; AI workload and external API usage should be considered as usage grows.